

IPSec VPN Setup on Ubuntu Linux Using StrongSwan (Complete Lab)

Lab Topology

VPN Server (Ubuntu)
Public/NAT IP : 192.168.1.132
LAN IP : 10.0.1.128

VPN Client (Ubuntu)
NAT IP : 192.168.1.133

Internal LAN
10.0.1.0/24

Goal:

Client
|
IPSec Tunnel
|
VPN Server
|
Internal LAN (10.0.1.0/24)

Step 1: Install StrongSwan

Server

```
sudo apt update
sudo apt install strongswan strongswan-pki -y
```

Client

```
sudo apt update
sudo apt install strongswan strongswan-pki -y
```

Step 2: Enable IP Forwarding on Server

Temporary:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Permanent:

```
echo "net.ipv4.ip_forward=1" | sudo tee -a /etc/sysctl.conf
sudo sysctl -p
```

Verify:

```
cat /proc/sys/net/ipv4/ip_forward
```

Output:

```
1
```

Step 3: Create IPsec Configuration on Server

Edit:

```
sudo vi /etc/ipsec.conf
```

Paste:

```
config setup

conn myvpn
    auto=add
    type=tunnel
    keyexchange=ikev2

    left=192.168.1.132
    leftid=@vpnserver
    leftsubnet=10.0.1.0/24

    right=%any
    rightid=@vpnclient

    authby=secret

    ike=aes256-sha256-modp2048!
    esp=aes256-sha256!

    dpdaction=restart
    dpddelay=30s
```

Save.

Step 4: Configure Pre-Shared Key on Server

Edit:

```
sudo vi /etc/ipsec.secrets
```

Add:

```
@vpnserver @vpnclient : PSK "cdac123"
```

Step 5: Configure IPsec Client

Edit:

```
sudo vi /etc/ipsec.conf
```

Paste:

```
config setup

conn myvpn
    auto=start
    type=tunnel
    keyexchange=ikev2

    left=%defaultroute
    leftid=@vpnclient

    right=192.168.1.132
    rightid=@vpnserver

    authby=secret

    ike=aes256-sha256-modp2048!
    esp=aes256-sha256!

    dpdaction=restart
```

Save.

Step 6: Configure Client Secret

Edit:

```
sudo vi /etc/ipsec.secrets
```

Add:

```
@vpnserv @vpncient : PSK "cdac123"
```

Step 7: Start StrongSwan

Server

```
sudo systemctl restart strongswan-starter  
sudo systemctl enable strongswan-starter
```

Client

```
sudo systemctl restart strongswan-starter  
sudo systemctl enable strongswan-starter
```

Step 8: Verify Tunnel

Server:

```
sudo ipsec status
```

Client:

```
sudo ipsec statusall
```

Expected:

ESTABLISHED

Step 9: Check Security Associations

```
sudo ip xfrm state
```

Expected output shows ESP security associations.

Step 10: Add Route to LAN (if required)

On Client:

```
sudo ip route add 10.0.1.0/24 via 192.168.1.132
```

(Depending on StrongSwan version/configuration this may be installed automatically.)

Step 11: Test Connectivity

Ping VPN server:

```
ping 192.168.1.132
```

Ping server LAN interface:

```
ping 10.0.1.128
```

Ping LAN machine:

```
ping 10.0.1.129
```

Important IPSec Ports

Allow these if a firewall exists:

Protocol	Port
IKE	UDP 500
NAT Traversal	UDP 4500
ESP	IP Protocol 50

Check Listening Ports

```
sudo ss -ulpn | grep -E '500|4500'
```

Expected:

```
*:500
*:4500
```

IPSec Packet Flow


